

EFY Note

The source code of this project is available for free download at source.efymag.com

ESPlorer software (Java based) is used to upload MicroPython application source code to NodeMCU. Steps to upload MicroPython program to NodeMCU through ESPlorer are:

- Start ESPlorer. Go to Settings and select MicroPython option.
- Select the Scripts option.

- Connect NodeMCU and select corresponding COM port, set baud rate to 115200.

- Open the port and press reset button on NodeMCU. Command prompt will display on the right as shown in Fig. 4.

- Open new file and write your program code, or open the Python script (*ReadTemperatureDisplayOn-Browser.py*)

After writing the above Python script, press 'Send to ESP' button as shown in Fig. 4. NodeMCU will be

Do-It-Yourself

connected to the Wi-Fi network. Note down the IP address 192.168.0.108 as shown in Fig. 5. Enter the same IP address in above Python program and press 'Send to ESP' button once again to upload the Python code to NodeMCU.

This project has also been successfully tested using Thonny IDE. The steps to use NodeMCU with ESPlorer and Thonny IDE are included in the source code folder.

Construction and testing

Connect temperature sensor LM35 with NodeMCU board with the help of three wires as shown in the circuit diagram (Fig. 2). You can also use the LM35 sensor available on a digital-analogue I/O module to test the project.

Open any web browser on PC (or smartphone) and enter the URL displayed on command prompt of ESPlorer as shown in Fig. 5. Ensure that the PC is connected to the same network access point where NodeMCU is connected.

If temperature is greater than 40°C, onboard LED of NodeMCU will glow and the circle displayed on web browser will become red. If temperature is less than or equal to 40°C, the LED will be off and the circle will become green. The screenshot of temperature display on a PC's web browser is shown in Fig. 6.

In the current source code, you need to press F5 (refresh) button on the PC to get new value of temperature on the browser. You can enter the same IP address on the smartphone and get the same temperature display.

Digital-analogue I/O module

The layout of a digital-analogue I/O (DAIO) module is shown in Fig. 7 and its schematic diagram is shown in Fig. 8. DAIO module has the following:

Digital inputs for any microcontroller. Four switches and infrared receiver TSOP1738

Analogue inputs for any microcontroller. Two variable resistors, one light sensor, and one temperature sensor

Digital and analogue outputs for any microcontroller. Eight LEDs and a buzzer

Three female-to-female connectors are required to connect LM35 sensor

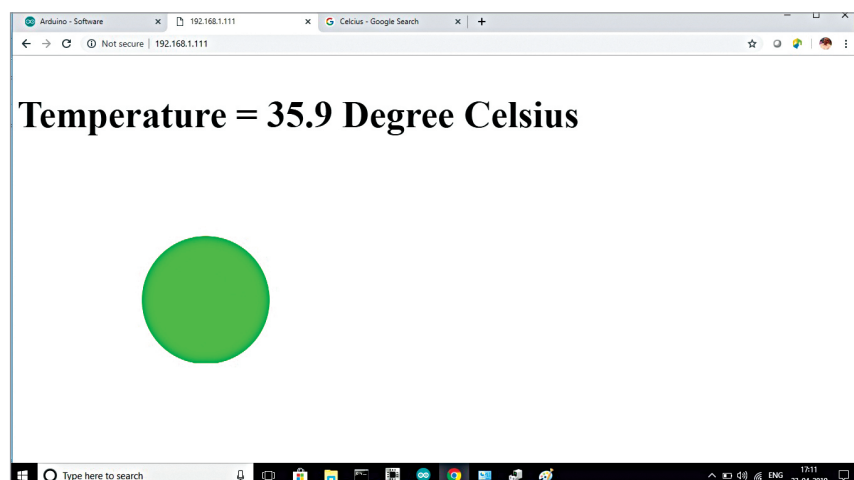


Fig. 6: Temperature display on web browser on PC

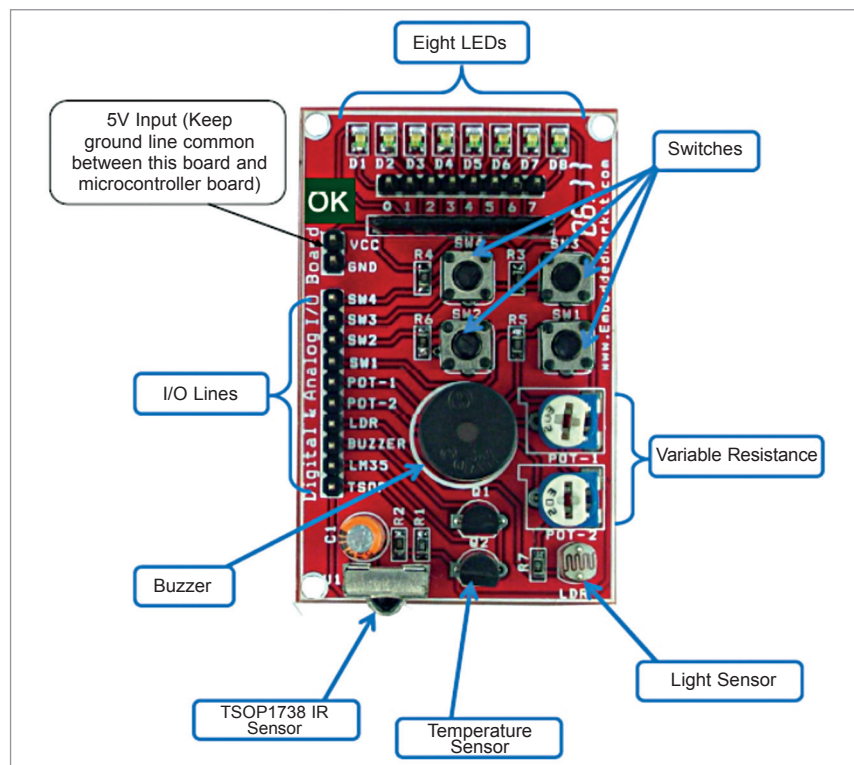


Fig. 7: DAIO board layout